REAGAN HU

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EDUCATION

University of Toronto - St. George Campus
Engineering Science - Bachelors of Applied Science and Engineering (Co-op) GPA: 3.8/4.0
University of Toronto Scholars Scholarship ($10000)

EXPERIENCE

AI Robotics Engineer Intern | *Acceleration Consortium* May 2026 – August 2026

- Engineered automated **Isaac Sim digital twins** allowing repeatable, high-volume data collection of **5000+** demonstrations for **imitation learning** and **reinforcement learning** enabling scalable policy training and investor-facing demonstrations.
- Established the first company-wide **single-operator** physical data collection workflow by developing an internal tool for voice-assisted teleoperation **(faster-whisper)** and automated segmentation for data annotation **(CV2, SAM2, PyTorch)**.
- Built an end-to-end inference pipeline (**WebSockets, PyZMQ, PyTorch**) enabling autonomous deployment of fine-tuned **VLA** policies supporting **DSRL** and **RTC** for multi-step workflows showcased to multi-billion-dollar venture capital firms.
- Increased precision autonomous manipulation accuracy by **75%** by developing an end-to-end, safety-gated visual-servo pipeline (**MolmoPoint, SciPy, ZED SDK**) for more reliable closed-loop robot alignment.

Autonomous Rover Embedded Engineer | *University of Toronto Robotics Association* September 2025 - April 2026

- Improved rover fault monitor execution throughput by **7.7x** through refactoring embedded **C++** codebase refactor into modular **FreeRTOS** task-based architecture with deterministic scheduling and **mutex**-protected shared variables.
- Reduced rover operation time by **67%** through building a **PyQt5** control GUI on **ROS-2 Humble**, implementing **publisher-subscriber** pipelines for live node/topic monitoring, launch control, and emergency stop handling.
- Developed a **Python** GPS/IMU analysis pipeline (**NumPy, Pandas, Matplotlib, PyProj**) transforming LLA to ENU coordinates with **Jacobian-based** covariance propagation for performance evaluation.

PROJECTS

AroundU
Skills & Tools: React, TypeScript, TailwindCSS, Vite, Motion, Node.js, Express, Supabase, OpenAI API, Hugging Face

- Built a full-stack campus social platform (**React**, **TypeScript**, **Vite**, **TailwindCSS**), implementing onboarding flows, real-time event discovery, messaging, and an interactive heatmap with **Google Maps API**.
- Developed reusable, animated interface components (**Motion**, **Lucide React**) supporting navigation and live visualization.
- Integrated a **Python** recommendation pipeline using **Hugging Face sentence-transformers** to embed and compare user profiles, with **OpenAI API** extracting and weighting data to generate compatibility scores.
- Contributed to backend development using **Node.js**, **Express**, and **Supabase PostgreSQL**, supporting user profiles, events, chats, and interest-based matching.

Fast and Flurrious Autonomous Rover
Skills & Tools: C++, Sensor Integration, PID control, PWM, Arduino

- Implemented a real-time line-following controller using calibrated **TCS3200 color-sensor** readings and tuned **PID control**, dynamically adjusting differential motor speeds through **PWM** for precise path tracking and line-loss recovery.
- Developed an **Arduino FreeRTOS** control architecture with concurrent sensor, motor-control, and mission-planning tasks, using **mutex**-protected shared state to coordinate autonomous rover behaviours.
- Integrated an **ultrasonic distance sensor**, **L298N motor driver**, and servo gripper for autonomous obstacle avoidance and object pickup and placement execution.

VOLUNTEERING

National Director of Design | *Hot Potato Initiative* September 2024 - June 2025
Skills: Marketing Design, Professional Collaboration, Project Management

- Coordinated a team of **10** in the development of digital and physical marketing materials for nationwide campaigns leading to **35%** outreach growth.
- Directed end-to-end creative workflows to develop national campaigns that balanced needs stakeholder needs with advocacy.
- Collaborated with executive, outreach, and advocacy teams to translate campaign objectives into cohesive visual strategies while balancing brand consistency, deadlines, and stakeholder feedback.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, Matlab
Tools & Libraries: Isaac Sim, CloudXR, SAM, PyTorch, TensorFlow, JAX, OpenCV, FreeRTOS, ROS-2, Pandas, Numpy
Skills: Forward/Inverse Kinematics, SSH, Digital Twins, Diffusion Models, Visual Servo
Coursework: CAD (Onshape), Circuit Design, Linear Algebra, Technical Writing
Languages: English, French, Mandarin